

Shock Problem and PINN Framework

1 Problem Setup

Consider the 1D inviscid Burgers Riemann problem.

Governing equation

$$u_t + uu_x = 0, \quad x \in [-1, 1], \quad t \in (0, 1]$$

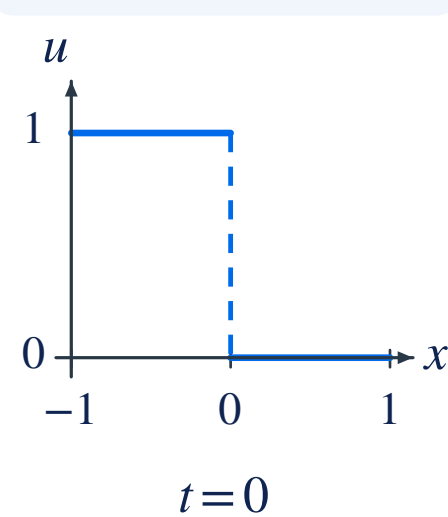
Initial condition

$$u(x, 0) = \begin{cases} 1, & x < 0, \\ 0, & x > 0. \end{cases}$$

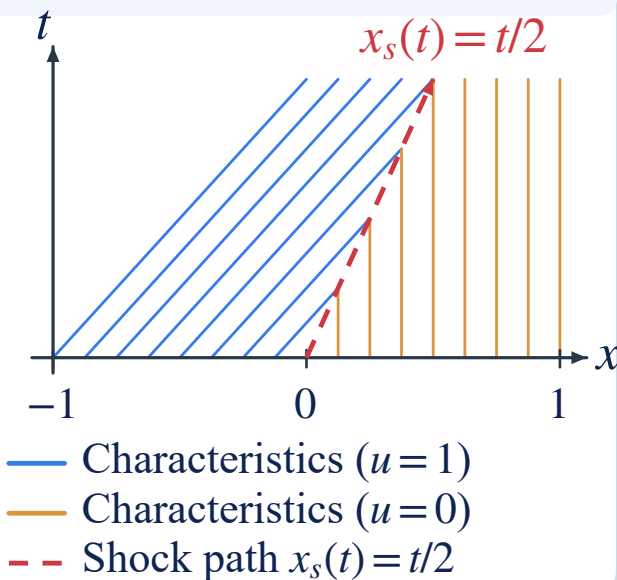
Boundary condition

$$u(-1, t) = 1, \quad u(1, t) = 0, \quad t \in (0, 1]$$

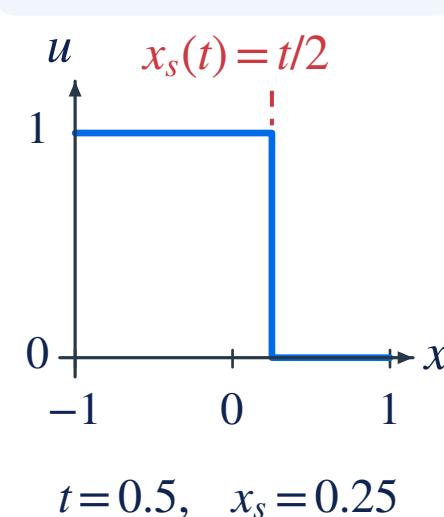
(a) Initial condition



(b) Characteristics



(c) Solution profile



- Characteristic speed: $dx/dt = u$.
- Characteristics converge toward the shock.
- Shock speed: $s = (u_L + u_R)/2 = 0.5$.

2 PINN Framework

